

1 · The question, and what this project set out to test

The Platonic Representation Hypothesis (Huh et al., 2024) in one sentence, the project's target in one sentence, and the difference between them.

The hypothesis

Models trained separately — different data, different objectives, sometimes different senses entirely — are said to be converging on the same underlying picture of the world. The name comes from Plato: there is one reality, and each model sees a shadow of it.

A CAT ON A CHAIR

DINOv2 sees pixels $\rightarrow x_{\text{img}} \in \mathbb{R}^{768}$

bge-m3 reads text $\rightarrow x_{\text{txt}} \in \mathbb{R}^{1024}$

$\text{cosine}(x_{\text{img}}, x_{\text{txt}}) \approx 0.00 \leftarrow$ the coordinates share NOTHING

and yet, in BOTH spaces:

$\text{dist}(\text{cat-on-chair}, \text{cat-on-sofa}) < \text{dist}(\text{cat-on-chair}, \text{airplane})$

That gap is the whole project. The numbers are unrelated; the relations are not.

What this project tests

The hypothesis as usually stated is not directly testable — 'converging on the same picture' has no units. This project narrows it to something that can be measured, and then measures it:

CLAIM UNDER TEST

independently trained models share representations UP TO A LINEAR TRANSFORMATION

which means: there exists ONE matrix W , fitted in closed form, such that

$$x_A \cdot W \approx x_B$$

no training, no gradients, no fine-tuning — just a solve

Three things make that a real test rather than a slogan. It is **falsifiable**: if no linear map works, the claim dies. It is **bounded**: a linear map is the weakest useful transformation, so success cannot be explained away as a powerful network learning the task by itself. And it is **useful**: if it holds, components from different models become interchangeable, which is an engineering result and not only a scientific one.

Two systems were built to test it

system	what it is	what it answers
Experiment B	two vision encoders, one 0.79 MB linear adapter, shipped as a Core ML artifact	Can two specific models be made interchangeable, well enough to deploy?
The Hub (Series G)	seven encoders, four training objectives, one 512-dimensional shared coordinate system	Can MANY models meet in one space, so a component trained on one serves the rest?

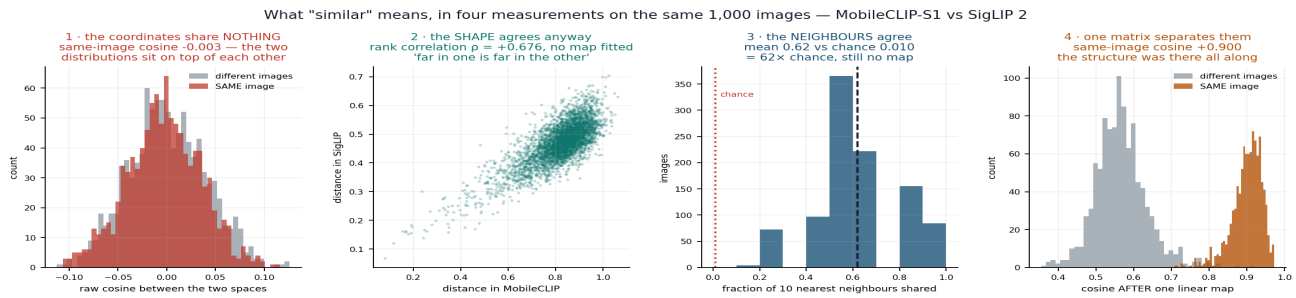
They answer different questions and carry different numbers. Keeping them separate matters: Experiment B's headline figure does not belong to the hub, and vice versa.

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2 · Five kinds of similarity, from weakest to strongest

"Are these two spaces similar?" is not one question. It is five, and they can have different answers for the same pair of models. Every claim in this project names which level it is about.



The same 1,000 images, four measurements. Panels 2 and 3 use NO fitted map — the agreement they show was already there.

level	the question, in plain terms	for the cat on a chair	how it is measured	result here
1 · shared NEIGHBOURS	Do the same things sit nearby?	in both spaces, is cat-on-sofa among its nearest items?	k-NN overlap, no map fitted	YES, strongly — hundreds of times chance at small k
2 · shared ORDER	Do they agree on what is nearer than what?	is cat-on-chair closer to cat-on-sofa than to an airplane, in both?	Spearman correlation of pairwise distances, no map	YES, moderately — ρ 0.29 cross-modal, up to 0.76
3 · shared STRUCTURE	Does the whole similarity pattern match?	does every item relate to every other in the same way?	CKA / RSA	PARTLY — 6.3x a random control, but weak in absolute terms
4 · shared SUBSPACE	Can one be turned into the other by a linear map?	can the image vector be mapped onto its caption's direction?	ridge regression, then retrieval	YES — 88–95% of a jointly-trained ceiling
5 · same GEOMETRY	Are they the same shape, just rotated?	does a pure rotation align the two clouds?	orthogonal Procrustes	NO — 0.038 against ridge 0.592. Related, not rigid

Reading the ladder

Each level is a stronger claim than the one above it, and the project's central result is that the answer **changes as you climb**. Neighbours agree strongly. Order agrees moderately. Global structure agrees weakly. A linear map works well. A pure rotation fails.

what that pattern means

strong at level 1, weak at level 3

→ the models agree about NEIGHBOURHOODS, not about global arrangement

level 4 works but level 5 fails

→ the spaces are related by a STRETCH, not merely a turn

This is why the project's conclusion is narrower than the hypothesis it started from. Models do not build the same space. They build different **charts** of one shared structure — like two map projections of the Earth, which disagree about distance and area while both keeping Spain next to France.

3 · The measurements, and how to read them

Every metric is a question with a blind spot. This page is the reference: what each one asks, when it lies, and what a high or low value actually means.

metric	what it asks	when it MISLEADS	how to read it
cosine	do two vectors point the same way?	in a collapsed space every pair reads +0.999 — the number is high and meaningless	always check the space's mean pair-cosine first
R ² (held-out)	do the predicted numbers match the target's?	if the target has tiny variance, R ² goes hugely negative even when retrieval is fine	reconstruction quality WITH magnitude; distrust on collapsed targets
R@1 / R@5	does the right item come first out of N?	meaningless without N — 0.967 of 10 and 0.557 of 1,000 are not comparable	the deployed question. Always quote against chance 1/N
% of ceiling	how much of a jointly-trained system survives?	if the ceiling was measured under a different protocol	the honest cross-experiment yardstick
k-NN overlap	are my neighbours your neighbours?	the raw fraction rises with k simply because chance does	read the RATIO to chance (k/N), never the raw number
CKA	do the two similarity structures match?	wider embeddings score higher even when independent	compare the RATIO to a random control, not the absolute
Procrustes	is it just a rotation?	—	low against a high ridge means anisotropy: a diagnosis, not a failure
effective rank	how many directions carry information?	—	below ~10% of width means collapsed; predicts almost everything else
pair-cosine	how alike are two RANDOM items?	—	near 0 healthy, +0.999 collapsed. The fastest single diagnostic
verification AUC	same item, or a random impostor?	the negatives are easy, so scores sit near 1	0.999 here means the QUESTION was easy, not that the model is perfect
rows per dimension	is the fit even powered?	below 5, a negative result may just be starvation	check BEFORE believing any failure

The lesson this table encodes

The project's recurring discovery is that **a metric is only interpretable relative to the geometry it is computed in**. The same system scored cosine 1.000 and retrieval 0.160 on the same data — not a contradiction, but two questions with different answers. Every surprising number in this project turned out to be a metric artifact or a geometry problem, never a broken pipeline.

Table C12, read in full

The published table gives R@1 and percent of native. It does not give the native R@1 itself, nor the rows behind each fit. Both are recoverable, and both change how the percentages read.

encoder / width	R@1 transferred	ratio (% of native)	calculation: native = R@1 / ratio	input rows -> rows per dim
DINOv2-small 768	0.398	111.5%	0.398 / 1.115 = 0.357 native	8,533 / 768 = 11.1 powered
DINOv2-base 1536	0.442	95.9%	0.442 / 0.959 = 0.461 native	8,533 / 1536 = 5.6 powered
DINOv2-large 2048	0.468	92.9%	0.468 / 0.929 = 0.504 native	8,533 / 2048 = 4.2 UNDER floor
ConvNeXt-base 1024	-	96.7%	held out; never in the hub	8,533 / 1024 = 8.3 powered
random-map control	0.000	-	chance = 1/1000 = 0.001	-

Three things to take from it. Absolute retrieval RISES with encoder size - 0.398, 0.442, 0.468 - while the percentage FALLS, because a bigger encoder supports a better native head and the denominator grows faster than the numerator. A falling percentage here is not falling performance. The recovered native figures match E1's separate direct image-to-bge ridge to three decimals on small and large (0.357 and 0.504), so the hub costs nothing at the reference point. And DINOv2-large is the one published fit below the project's own floor of 5 rows per dimension - which biases ridge DOWNWARD, making 92.9 percent conservative rather than flattering. Report E.13.

4 · Experiment B — two encoders, one matrix, 0.79 MB

The narrowest version of the claim, tested to deployment. A phone-sized vision encoder and a server-sized one, never trained together, made interchangeable by a single linear map.

THE SETUP

MobileCLIP-S1 (phone tier) → 512 numbers per image
SigLIP 2 (server tier) → 768 numbers per image

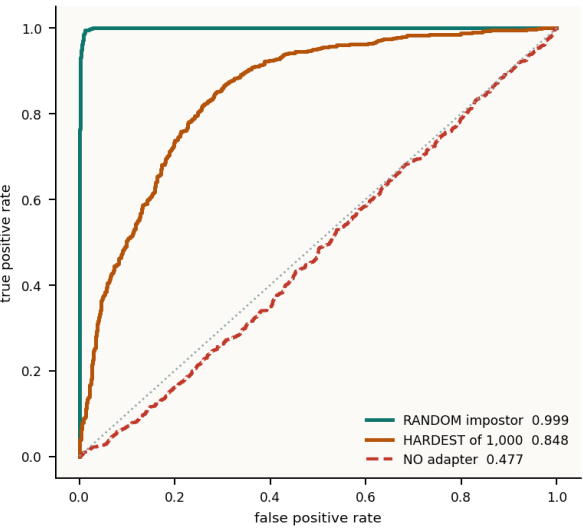
THE FIT — closed form, no training

$$W = (X^T X + \alpha I)^{-1} X^T Y \quad \text{on 3,000 image pairs}$$

THE TEST — on 1,000 images the fit never saw

phone vector * W → does it retrieve the SAME photo?

Experiment B · MobileCLIP-S1 → SigLIP 2
0.79 MB adapter, 1,000 held-out



Why AUC 0.999 is not overfitting

not overfitting — held out

fitted on 3,000 pairs, scored on 1,000 unseen. Overfitting would inflate every score; it does not.

the same map, three difficulties

vs a random impostor 0.999
vs the hardest of 1,000 0.848
single-best R@1 0.987
A memorised map does not fall 0.15 on harder negatives.

0.999 = an easy question

"same image, or a random other one?" — usually a dog against a kitchen. Near-ceiling is expected here.

quote all three, never 0.999 alone

0.999 typical impostor · 0.848 hardest negative · 0.987 one-of-a-thousand. One system, three questions.

control

remove the adapter: AUC 0.477, chance. Everything above is the 0.79 MB matrix.

The same adapter, asked three questions of increasing difficulty. The spread is what proves it is not memorisation.

measurement	result	what it means
raw, no adapter	R@1 0.001 = exact chance	the two spaces share nothing usable in their own coordinates
ridge adapter	R ² 0.592, cosine 0.900	one matrix recovers most of the relationship
MLP instead of a matrix	the network FAILED to beat the matrix — gain +0.0000	given more capacity, non-linearity and 38,000 rows, 40 epochs of training arrived at exactly the number a closed-form solve produces. Not 'a linear map is good enough' — the relationship IS linear
retrieval	R@5 95.3% of a jointly-trained ceiling	good enough to deploy
Procrustes (rotation only)	R ² 0.038 vs ridge 0.592	a pure rotation fails — the spaces differ in stretch as well as orientation
shipped artifact	0.79 MB, fp16, iOS 16+	a phone can hold the entire translation between two model families

One caption or five? A decision, not an oversight.

Experiment B (the phone adapter) embeds ONE caption per image; the hub's E-series averages all FIVE. The difference is deliberate and follows from what each measures.

The hub needs an ANCHOR. Its text vector must line up row-for-row with the image vector, so it averages the five captions into one stable, denoised summary per image. (The report confirms this is not load-bearing: the hub verdict was run both ways and lands in the same band.)

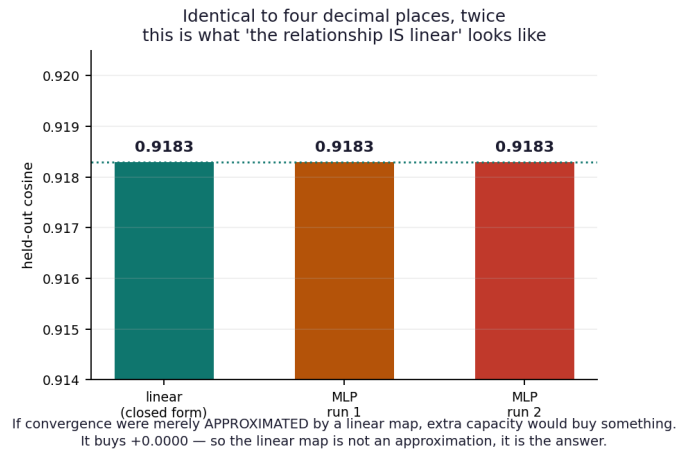
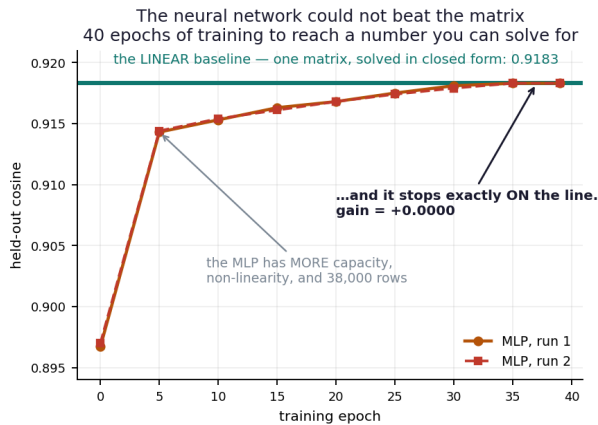
Experiment B measures a QUERY. B3 is a deployment test - a user types a phrase and the right photo should return. A real query is ONE caption, not an average of five the user does not have. Averaging would RAISE the score by measuring a query that can never be issued, so single-caption is the honest metric here, and - like the anisotropic target - it makes the reported number conservative rather than flattering.

Note: averaging is a TEXT-side operation only. Image vectors are never averaged across images - the cls+patch pooling averages patch tokens WITHIN one image, which is a different operation. Ref: report C.9 caption-protocol note.

5 · The strongest evidence in the project is a failure

Finding 1 is usually stated as "convergence is linear". That undersells it. The result is that a neural network, given every advantage, could not beat a matrix.

The contest. A neural network was given every advantage over the matrix — non-linearity, far more parameters, twelve times more training data, forty epochs of gradient descent — and it could not beat it. It did not come close and fall short; it landed on exactly the same number, 0.9183, to four decimal places, on two independent runs.



Real training traces from both runs. The teal line is what one matrix achieves with no training at all.

Why that failure is the strongest evidence in the project: if convergence were merely APPROXIMATED by a linear map, extra capacity would buy something — a little, or a lot, but something. It bought +0.0000. A linear map is therefore not a convenient approximation of the relationship between these two models. It is the relationship.

6 · The Hub — seven experts, one meeting place

Experiment B connects two models. Real systems have many. Connecting them pairwise does not scale; a shared coordinate system does.

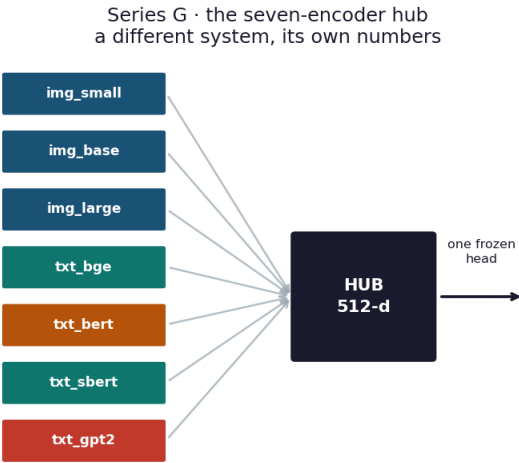
THE PROBLEM WITH PAIRS

7 encoders, connected pairwise → $7 \times 6 = 42$ maps
add one more model → 14 MORE maps to fit

THE HUB

7 encoders, each with one map INTO a shared space → 7 maps
add one more model → 1 map, and nothing else changes

N maps instead of $N(N-1)$



N maps, not N^2 — a new encoder is ONE closed-form solve

Reading the hub's numbers

NOT the Experiment B number

0.999 belongs to TWO encoders and a shipped adapter.
The hub serves seven and carries its own figures.

what the hub measures

transfer AUC 0.986 (G1, seven spaces)
transfer R@1 0.451 at HUB_DIM 512
pairs sharing one space 42 / 42 at 103.6% of bespoke
component transfer 92.9-96.5% of a native head

why lower than 0.999

a harder question: do SEVEN encoders with four training
objectives land close enough in one coordinate system to
be read by a component none of them trained with?

controls

random map in the hub slot → chance (0.004 vs 0.001)
cross-lineage SigLIP 2 → 94.2%, inside the 93.8-96.5%
within-family band. Ancestry was not doing the work.

the bound

one image domain, one language. Reads out of every space;
writes into all but collapsed GPT-2 (24.8%).

The hub is not a trained network — it is a whitened PCA plus one ridge solve per encoder. Nothing to maintain, nothing to retrain.

Why many small experts beat one large generalist

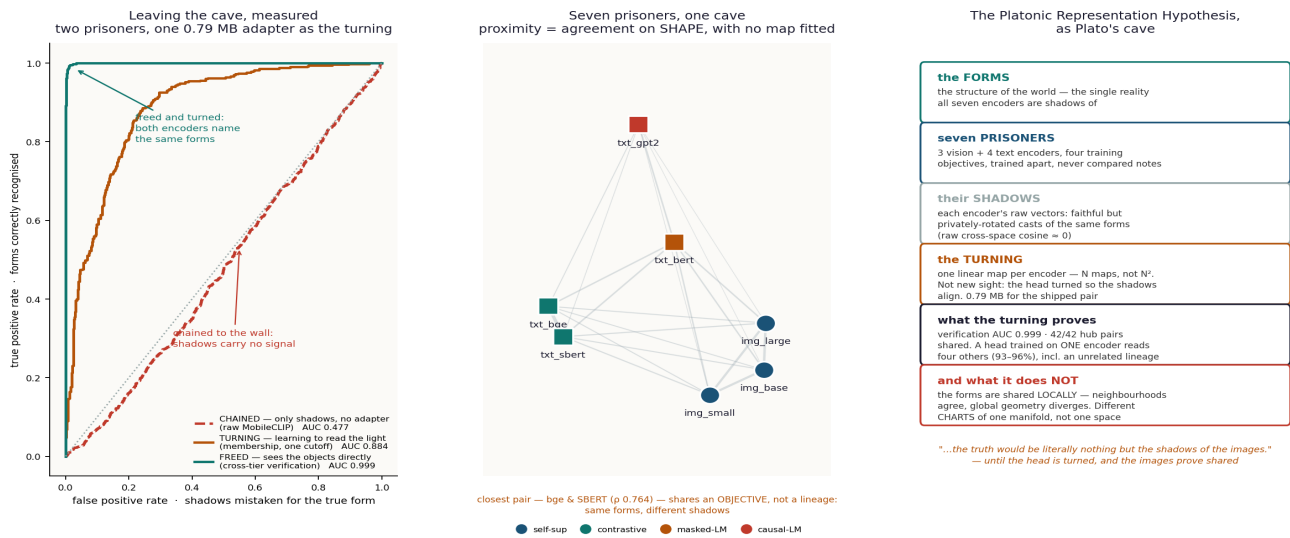
property	one large model	seven experts + hub
adding a capability	retrain or fine-tune the whole model	fit ONE new map; every existing map untouched
replacing a component	risky — everything depends on it	swap one arm, refit its map alone
cost of an update	full training run	a closed-form solve, seconds
specialisation	one model averaged over all domains	each expert strong in its own domain
measured cost of sharing	—	103.6% of hubs built for each pair alone — sharing is FREE
measured benefit	—	a component trained on ONE encoder works on encoders it never saw, at 92.9–96.5% of native

The honest cost, since closed. Hub width looked like a trade-off: transfer rises to 0.489 at 512 dimensions and falls to 0.312 at 768. It is not one. The cliff is least squares dividing by a near-null singular value, which appears exactly when hub width reaches the source encoder's rank - a numerical artifact of the fit, not a property of shared coordinates. The rule is $\text{hub_width} < \text{dim}(\text{source encoder})$, and a truncated or regularised head lifts the constraint entirely (report, E.8).

One GPU pass, then a ruler. The only step that needs a GPU is the one-time encoding pass that builds the caches · the eight encoders run forward over 9,533 items once, on Colab (T4/L4). Everything after is closed-form linear algebra on CPU: the ridge maps, the hub, all transfer. The expensive hardware makes the raw material once; the science runs without it · the same economy as the 0.79 MB adapter that ships without a GPU.

7 · The cave — what the allegory does and does not license

Plato's prisoners see shadows on a wall and take them for reality. The analogy is genuinely useful here, and it is also the place where this project's claim is most easily overstated.



Plato	this project	the cat on a chair
the FORMS — one reality	the structure of the world that all encoders are trained on	the actual fact that cats sit on chairs, and that this resembles cats on sofas
the PRISONERS	seven encoders, trained apart, never compared notes	DINOv2 saw the photo. bge read the caption. Neither saw the other.
the SHADOWS	each encoder's raw vectors	x_img and x_txt: faithful casts of the same fact, privately rotated
the TURNING	one linear map per encoder	not new sight — the head turned so both shadows line up
walking out	this is where the analogy STOPS	no encoder ever sees the form directly. They agree with each other; that is all we measured

Where the allegory misleads

Plato's prisoner turns and sees **one** object casting **one** shadow. That would predict that two encoders agree about everything, at every scale. They do not. Agreement is strong for near neighbours and fades to almost nothing across the whole space — so the honest picture is not one object and one shadow, but **two different projections of the same territory**, each faithful locally and distorted globally.

the claim the evidence REFUTES

"good models converge to the same embedding space"
→ raw cosine ≈ 0 , global agreement $\approx$ chance

the claim the evidence SUPPORTS

"models are different CHARTS of one shared relational structure"
→ coordinates differ entirely; local relations survive the change

8 · What was found, proved, and discovered

Nine findings, each held-out and each quoted against a control, plus the two discoveries that were not being looked for.

#	finding	the evidence
1	Convergence is linear — proved by a failure	a neural network with more capacity, non-linearity and 38,000 training rows COULD NOT BEAT a closed-form matrix. Not narrowly: 0.9183 against 0.9183, gain +0.0000 , on two independent runs. Extra capacity buys nothing, so the linear map is not an approximation of the relationship — it IS the relationship
2	Related, not rigid	a rotation-only map scores 0.038 where a full linear map scores 0.592 — a 15× gap. The map's singular values span 69×, so the spaces differ in STRETCH along different directions, not merely in orientation
3	Deployable	95% of a jointly-trained ceiling from a 0.79 MB fp16 matrix; verification AUC 0.999
4	It crosses modality	a vision encoder that never saw a word of text reaches 40.9–58.3% of a jointly-trained ceiling inside a text space — up to 358× chance . Nothing in either training run rewarded agreement with the other
5	It grows with capacity	+15.3 points of ceiling per decade of parameters, six monotone measures, over a 13.8× range
6	Isotropy, not similarity training	whitening a raw language model's space — label-free, linear — reaches 102.6% of an encoder trained contrastively on 1.2 billion pairs
7	The correspondence identifies itself	Gromov-Wasserstein recovers 39.7% of pairings exactly from shape alone, 595× chance, with zero paired examples
8	One shared space, actually used	42 of 42 encoder pairs share one 512-d hub at 103.6% of bespoke; a head transfers to unseen encoders at 92.9–96.5%, including an unrelated lineage at 94.2%
9	The mechanism predicts	a pre-registered prediction about two unseen encoders confirmed at rank correlation +1.00

Two things discovered rather than sought

discovery	what it says	why it matters
The sharing is LOCAL	neighbourhood agreement runs hundreds of times chance; global agreement falls to about twice chance. Confirmed on three separate metrics, including the one the original authors designed	it narrows the Platonic claim to its defensible form, and it explains why a linear map reaches most but not all of the ceiling
Isotropy has two opposite effects	the same geometric property HELPS reading a known correspondence and HURTS finding an unknown one - measured twice. The cliff was once a third instance; it is rank, not isotropy (E.8)	one property, opposite prescriptions: there is no single 'good geometry', only geometry good for a stated task

And the methodological result. Ten explanations offered during this work were tested and recorded as false - two of them the project's own, and one entered on a faulty test and formally withdrawn - rather than quietly dropped. A claim that survived a test which could have killed it is worth more than one never exposed to one.

9 · Caveats, and what could be done next

Every claim above has a boundary. Naming them is not modesty — an unbounded claim cannot be tested, and a reviewer will find the boundary anyway.

What the results do NOT say

claim	measured scope	NOT claimed
convergence is linear	GPT-2/Pythia; MobileCLIP/SigLIP; DINOv2 ×3; four text encoders	all architectures, all scales
alignment grows with capacity	DINOv2 family, 22M–304M parameters	behaviour beyond 304M, or in other families
isotropy rescue	one collapsed target, two image scales	that isotropy is SUFFICIENT — it is necessary only
pair-free recovery	39.7%; both encoders contrastive image-text	a deployable system that needs no pairs
the shared hub	7 transformer encoders, 4 objectives, COCO, English; held-out SigLIP 2 and ConvNeXt	a universal space for all models, domains or languages
writing into a space	reads out of all seven; writes into all but collapsed GPT-2 (24.8%)	that a collapsed space can be reconstructed for ranking
structure is shared	LOCALLY — neighbourhoods agree, global geometry diverges	global metric identity
domain	one image domain, one language, one caption corpus	open-world scale, other domains or languages

What could be done next, in order of value

next test	the question	cost	why it matters
A convolutional encoder	every encoder here was a transformer. RUN, and closed: ConvNeXt-base joins at 96.7% of native	ran — control at chance (0.001)	ABOVE the within-family band (93.8–96.5%); architecture does not bound the claim
Experts vs one large model	do K small domain experts in a shared hub match one big general encoder, at matched parameters?	medium	turns the engineering argument into a benchmark someone outside this field would care about
More encoders per group	shared OBJECTIVE predicts agreement better than shared lineage — but on as few as one pair per cell	medium	the most interesting causal hint in the project, and currently the least powered
Other languages and domains	does any of this survive outside COCO captions in English?	high	the broadest of the stated limits
Non-linear hubs	a neighbourhood-preserving embedding might carry local structure where whitened PCA does not	medium	ruled out here by the no-training constraint, but it is the natural sequel
Reverse-direction transfer	does the hub carry information recoverable TOWARD an image space, not just FROM one? Source/target roles are independent (G10, SBERT) — untested for images	medium	first image-target test; ConvNeXt is the cleanest case since it was never a hub member, so no overlap question to declare. Mirrors G10's source-composition sweep, applied to which encoder trains the head
Scale the text side	PRH's own capacity ladder predicts alignment rises with encoder size — tested here only up to bge/BERT/GPT-2 scale (max 1024-d). Does a 7B+ text encoder push transfer and shape agreement higher, as the ladder predicts?	medium	the closed-form design means adding a larger encoder costs one entry-map solve, same as any other. The scale caveat an examiner is most likely to raise — answered in advance if this runs before the defense, conceded cleanly if not
Re-estimate the hub on a larger corpus	The hub PCA is estimated on 8,533 rows over a 5,376-dim concatenation — aspect ratio $\gamma = d/n = 0.63$, and 0.90 for the seven-space version at 7,680 dims. Full rank, but near $\gamma = 1$ the sample spectrum distorts. Do the transfer figures hold, and does DINOv2-large's 92.9 percent rise, at a lower γ ?	high	the only change that improves the FOUNDATION rather than adding a rung. A 35,133-item corpus brings γ to 0.22 and lifts img_large's entry map from 4.2 to 16.7 rows/dim, with eval precision tightening from ± 3.1 to ± 1.6 points. Mitigating now: only the top 512 directions are kept, the best-estimated part of the spectrum, and the residual bias runs conservative. Every published number would need re-measuring — a post-submission job

What could be done next (continued)

Whiten GPT-2 before it enters the hub	C.11 rescues GPT-2 by whitening its own 768-d space in full - rank 6.5 to 490.8, R@1 0.130 to 0.478. The hub never applies that: it scalar-rescales each space, concatenates, truncates to 512, and whitens only the survivors. Is a rescued GPT-2 worth more to the hub than a collapsed one?	low a one-line change in G1 and a rebuild. G10 already shows GPT-2's presence is worth several points (90.3 vs 85.5 images-only), so the space contributes despite its collapse. The scalar rescale is deliberate and correct for its own purpose - per-column standardisation would inflate near-dead columns before the PCA, the E.8 failure mode - so this tests a third option neither currently covers
Rebuild the hub WITH DINOv2-giant as a member	Giant is currently a SOURCE: it gets an entry map into the frozen hub and the head reads its coordinates, which is what makes its result a held-out test. As a MEMBER its 3,072 dimensions would join the concatenation and reshape the basis itself. Does a hub built around a much wider, higher-capacity image encoder transfer better - given that giant's RAW agreement fell?	blocked until the corpus grows adding giant takes the seven-space concat to 10,752 dims against 8,533 training rows: gamma = 1.26, so the covariance is RANK-DEFICIENT and the basis is not well-posed at all. (ConvNeXt was already blocked at 1.02.) At 35,133 items gamma falls to 0.32 and it becomes feasible - so this DEPENDS on the row above it. Note the trade: as a member, giant stops being a held-out test, and every published figure is re-measured

THE PROJECT IN ONE PARAGRAPH

Two models never share coordinates · raw agreement is exactly chance. But they describe the same world, and the translation between them is a single matrix small enough to fit on a phone. What they share most strongly is LOCAL: they agree about what is near what, and diverge about the shape of the whole. The coordinate frame that makes structure readable is cheap; the arrangement of meaning across directions is what training actually buys.

11 · The K-series and H1 · robustness, reproduction, and power

Why these notebooks exist. The G-series produced all the headline results, but three questions about those results needed separate, independent tests: are the raw-space comparisons in C.13 replicable after the VM died? Is the 'no detectable effect' in the local-structure test a real null or a starved measurement? And how do we know any negative result is powered enough to believe? K1-K3 answer the first two; H1 answers the third.

K-SERIES · reproduction and bounding the local-structure null

K1 · Raw Procrustes across all 21 pairs (tests C.13.7). Runs Procrustes alignment in the raw embedding spaces · no hub · for all 21 encoder pairs, using the bge-SBERT anchor pair as an alignment check (152.8× chance confirms the id bookkeeping). It fits a global scale with the rotation, whereas G-series used scale-free Procrustes. Produces the raw-space structural comparison that C.13.7 references.

K2 · G7 re-run · per-pair kNN gains (tests C.13.10). G7's per-pair local-structure measurements died with the VM. K2 regenerates them from the cached encoder vectors. It gates on the healthy/degenerate PATTERN reproducing (healthy pairs near zero, degenerate ones clearly positive, Spearman > 0.4), not on exact agreement with the original numbers. Writes g7_rebuilt.npz, which K3 reads.

K3 · Bounded null · from absent effect to a measured bound (tests C.13.10). Reads g7_rebuilt.npz from K2. The healthy pairs are all combinations of four encoders · six pairs, not independent · so the primary interval is a cluster bootstrap over encoders rather than a naive t-interval. Pre-registered SESOI (smallest effect of interest) = 0.050. If the 95% CI fits inside ± 0.050 , the result is upgraded from 'no detectable effect' to 'bounded within ± 0.05 ', which is a far stronger claim: not 'we couldn't see it' but 'it cannot be large'.

H1 · Power diagnostics: run before believing any negative result

What it catches. Any experiment that produces a null · no effect, no detectable difference · is only credible if the fit was powered to detect an effect of the relevant size. H1 checks four things before you trust a negative: (1) artifact check · NaN, zero rows, un-normalised vectors; (2) geometry check (UNCENTRED effective rank and mean pair-cosine, so collapse is not hidden by centring); (3) power check · rows per input dimension ≥ 5 floor; (4) hub-artifact check · hub PCA aspect ratio $\gamma = d/n$ and entry-map rows/dim for each encoder.

What it cannot decide. H1 separates broken outputs from healthy ones, but it cannot distinguish (a) a genuinely anisotropic space from (b) wrong pooling that mimics collapse. A causal model fine-tuned for embeddings that is mean-pooled instead of last-token pooled produces the same low-effective-rank signature as GPT-2's real collapse. When the geometry check fires on an embedding-fine-tuned model, verify the pooling FIRST before reaching for whitening.

Relationship to the K-series. H1 is the meta-test that justifies running K3: before claiming a bounded null, check that the underlying measurement was powered. K3 reads g7_rebuilt.npz; H1 checks g7_rebuilt.npz's geometry and power before K3 runs. H1 is therefore a prerequisite, not an alternative.

10 · The fractal probe — is it the *data*? (G8–G10, post-defense)

PRH says shared structure comes from training on the same reality. The sharpest way to test that clause is to train on a reality that contains none of the natural world — pure fractals — and ask how much shared structure survives. Three notebooks added after the defense do this. They continue as ongoing research; the defended claims above are unchanged.

The probe

Two vision models trained **only** on FractalDB-1k (one million rendered fractals, a thousand formula-defined classes, zero natural images) — a ResNet-50 and a DeiT-tiny — are measured against all eight roster encoders by the same map-free shape agreement (level 2 of the ladder). Each is bracketed by a same-architecture **random-init** twin (the floor of ‘architecture only’) and an **ImageNet-1k** twin (the ceiling of ‘architecture + real images’), with a 32×32 raw-pixel floor beneath. The reality fraction $R = (\rho_{\text{fractal}} - \rho_{\text{random}}) / (\rho_{\text{natural}} - \rho_{\text{random}})$ places the result on a 0-to-1 scale between those brackets. The pipeline reproduces the published ConvNeXt row and bge–SBERT 0.768 to the decimal before any fractal number is trusted.

What the fractal models share (G9)

prediction	ResNet-50	DeiT	reading
fractal < ImageNet twin	confirmed 8/8	confirmed 8/8	fractals fall short of real images
fractal > random twin	confirmed 4/4	confirmed 4/4	training adds real structure
image-vs-text R gap	inconclusive	inconclusive	no reality-dependence asymmetry
fractal > pixels (raw)	falsified 0/4	confirmed 4/4	CNN sits at the pixel floor

Reality fraction $R \approx 0.09$ for the CNN and **0.21** for the ViT: fractal-only training installs roughly a tenth to a fifth of what natural images give. The ViT shows it directly. The CNN’s raw agreement sits at the pixel floor — not because it shares nothing, but because its space is **anisotropic** (mean pair-cosine 0.85, effective rank ~368). After held-out de-anisotropization the fractal CNN reliably clears the whitened controls ($R_{\text{white}} \approx 11\%$). This is the E.6/E.12 lesson a third time: raw geometry on a degenerate space measures the post-collapse residual, and whitening *exposes* a small real signal rather than creating one. The image-vs-text comparison (P3) is **inconclusive** under an item-level paired bootstrap — the earlier ‘confirmed’ reading was a bootstrap artifact and was withdrawn.

The preprocessing control

The random CNN twin used ImageNet normalization while the fractal CNN used the FractalDB repo norm, so ‘fractal > random’ risked conflating training with preprocessing. A matched-norm control — a random ResNet-50 under the **FractalDB** norm — settles it: random rises from 0.032 (ImageNet norm) to 0.036 (FractalDB norm), while the fractal model sits at 0.067. The norm accounts for about a tenth of the apparent gap; the training effect survives on the matched baseline (0.067 vs 0.036). The claim holds, corrected.

Is that structure *recoverable*? (G10)

Shape agreement measures what is visible without fitting. The hub test asks the stronger question: does a fitted linear entry map carry the fractal features into the frozen 512-d hub such that the frozen caption head — trained on DINOv2-small, never on fractals — can read them out? Same protocol as the cross-lineage and ConvNeXt transfers; the reproduction gate passes first (img_base 94.8%, img_large 93.4% of native).

encoder	transfer R@1	native R@1	random twin	verdict
fractal CNN	0.014	0.014	0.012	at floor — tied with random
fractal ViT	0.020	0.022	0.017	at floor — tied with random
ImageNet CNN	0.258	0.274	—	clears the floor
ImageNet ViT	0.204	0.208	—	clears the floor

Fractal transfer sits **at the floor**, indistinguishable from its random twin. ‘Percentage of native’ reads ~100% only because the native head is *also* at the floor — a ratio of two near-zero numbers, which the notebook guards out as **not meaningful** rather than reporting as success. The natural-image encoders clear the floor at 0.20–0.26 and prove the protocol itself works. The precise statement: under the frozen hub/head protocol, FractalDB

encoders show **no demonstrated training-specific transfer advantage beyond random initialization**.

What the probe adds to the picture

In one line: structured synthetic training creates a small amount of *visible* shared geometry — more in the transformer, and in the CNN only after de-anisotropization — but natural-image training creates much more, and that weak fractal geometry does **not** become a training-specific advantage in semantic caption retrieval through the shared hub. This is a weak, data-driven reading of PRH: natural-world training is far more effective at producing structure that transfers into a shared semantic function. It does not claim a universal representation, and it finds no image-versus-language difference in how much reality the shared structure needs.

Scope. These are 768–2048-d encoders and two released fractal checkpoints (the CNN is a ResNet-50, the ViT a DeiT-tiny — a capacity gap that keeps the CNN-vs-ViT contrast uninterpreted). The finding bounds fractal-only training at this scale; it does not test the FDSL literature’s larger models.

Four implications for how alignment is measured

The project’s cumulative results carry four implications beyond the specific claim, each earned by a named finding rather than offered as reflection.

1 · Check isotropy before believing a failure. Anisotropic spaces make cosine and kernel metrics report an absence of alignment that a three-line whitening removes. A reported non-convergence may describe the coordinate frame, not the content. (GPT-2, the fractal CNN, ConvNeXt.)

2 · Visible is not recoverable. A low RSA/CKA/p is not evidence that two models cannot be aligned — ConvNeXt has the lowest raw agreement and the highest hub transfer. But when structure is genuinely absent, a fitted map correctly finds nothing (the fractal encoders at the floor). Measure both; raw geometry alone does not tell you whether alignment is possible.

3 · There is no single ‘good geometry’. Whitening helps *reading* a known correspondence (+102.6%) and destroys *finding* an unknown one (39.7% → ~0). Any advice to whiten, or not, is incomplete until it names the task.

4 · Convergence is local. Neighbours agree hundreds of times chance; global structure only about twice. ‘Models converge on the same representation’ is defensible as convergence on the same local neighbourhood structure — not a shared global geometry.

The models, by type and pooling

vision model	type	params ≈	width	training signal
MobileCLIP-S1	conv/transf.	21.5M vis.	512	image–text pairs + distillation
SigLIP 2 base	ViT	0.4B	768	image–text sigmoid + aux.
DINOv2 S/B/L	ViT	22/87/304M	768/1536/2048	self-supervised views, no captions
ConvNeXt-base	ConvNet	89M	1024	supervised ImageNet-22k
DINOv2 giant	ViT	1.14B	3072	visual self-supervision (larger)
FractalDB ResNet-50	ConvNet	~25M	2048	FractalDB-1k only
FractalDB DeiT-tiny	ViT	~5M	384	FractalDB-1k only

text model	type	params ≈	width	training / role
GPT-2	transf. (causal)	124M	768	next-token / Exp A + hub text
Pythia-160M	transf. (causal)	160M	768/layer	next-token / controls
BERT-base	transf. (masked)	110M	768	masked tokens / hub member
SBERT all-mpnet	transf. (bi-encoder)	110M	768	sentence-pair contrastive / hub
bge-m3	transf. (bi-encoder)	568M	1024	retrieval / hub + caption target

Appendix G — Resources and links

All links verified as of September 2026. Where a resource is available from multiple sources, the canonical or original publication is listed first.

Papers

resource	use in this project	link
Huh, Cheung, Wang & Isola (2024) Platonic Representation Hypothesis	scientific motivation, the claim tested	ICML/PMLR 235 arxiv.org/abs/2405.07987
Kornblith, Norouzi, Lee & Hinton (2019) Similarity of Neural Network Representations Revisited	JCA metric and debiased variant	ICML/PMLR 97 arxiv.org/abs/1905.00414
Kataoka et al. (2020/2022) Pre-training without Natural Images	FractalDB-1k dataset and pretrained checkpoints (G9/G10)	ACCV 2020 / IJCV 2022 arxiv.org/abs/2101.08515

Datasets

dataset	use	link
COCO 2017 (train2017)	9,533 image-caption pairs, all encoder caches	cocodataset.org
Salesforce WikiText-103	GPT-2 / Pythia passages (Experiment A)	huggingface.co/datasets/ Salesforce/wikitext

Vision models

model	source	link
Apple MobileCLIP-S1	image-text pairs + distillation	huggingface.co/apple/ MobileCLIP-S2-OpenCLIP
Google SigLIP 2 base	image-text sigmoid loss	huggingface.co/google/ siglip2-base-patch16-224
Meta DINOv2 S/B/L/G	self-supervised visual views	github.com/facebookresearch/dinov2
ConvNeXt-base-22k	supervised ImageNet-22k	huggingface.co/facebook/ convnext-base-224-22k
FractalDB ResNet-50	FractalDB-1k pre-training	github.com/hirokatsukataoka16/ FractalDB-Pretrained-ResNet-PyTorch
FractalDB DeiT-tiny	FractalDB-1k pre-training	github.com/hirokatsukataoka16/ FractalDB-Pretrained-ViT-PyTorch

Language and text models

model	source	link
OpenAI GPT-2 (124M)	next-token prediction	huggingface.co/openai-community/gpt2
EleutherAI Pythia (160M)	next-token prediction	github.com/EleutherAI/pythia
BERT-base (110M)	masked language modelling	huggingface.co/google-bert/ bert-base-uncased
SBERT all-mpnet-base-v2	sentence-pair contrastive	huggingface.co/sentence-transformers/ all-mpnet-base-v2
BAAI bge-m3 (568M)	multilingual retrieval	huggingface.co/BAAI/bge-m3

Software and infrastructure

tool	use in project	link
PyTorch	model loading, feature extraction	pytorch.org
Hugging Face Transformers	tokenizers, model API	huggingface.co/docs/transformers
timm	ViT/DeiT model loading	github.com/huggingface/pytorch-image-models

scikit-learn	ridge regression, PCA	scikit-learn.org
Apple Core ML Tools	fp16 adapter export to iOS	github.com/apple/coremltools
Qdrant	shared retrieval index	qdrant.tech